🌐 📘 Chapter 33-EtherChannel (Port Channel) 🚀

🧠 What is EtherChannel?

👉 **EtherChannel (also called Port Channel / LAG - Link Aggregation Group)** allows you to combine multiple physical interfaces into **one logical interface**.

💡 **Simple Definition:**

- Multiple cables → act like **ONE cable**
- Multiple ports → behave like **ONE port**

🎯 Types of EtherChannel

🔹 **1. Layer 2 EtherChannel**

- Group of **switch ports**
- Works like a **single Layer 2 interface**
- Used for switching (VLANs, trunks)

🔹 **2. Layer 3 EtherChannel**

- Group of **routed ports**
- Works like a **single Layer 3 interface**
- You can assign an **IP address** directly

🏠 Real-Life Example

Imagine a **highway** 🛣️:

- One road = slow traffic 🚗
- Multiple lanes = faster traffic 🚗 🚗 🚗

👉 EtherChannel = **multiple lanes combined as one big road**
👉 More speed + no traffic jam = better performance 🚀

⚠️ Problem Without EtherChannel

🔸 **Oversubscription**

- End devices bandwidth > switch uplink bandwidth
- Leads to **congestion** 😓

🔸 **STP Issue**

- If multiple links exist:
  - STP blocks all except ONE ❌
  - Only one link is used 😑

🔥 **Why?**

- To prevent **Layer 2 loops**
- Otherwise → **Broadcast storm** 💥

✅ **Solution: EtherChannel**

✔ All links are ACTIVE (no blocking)
✔ STP treats them as ONE interface
✔ Traffic is **load balanced**
✔ No loops

```
ASW1(config-if-range)#channel-group 1 mode ?
  active      Enable LACP unconditionally
  auto        Enable PAgP only if a PAgP device is detected
  desirable   Enable PAgP unconditionally
  on          Enable Etherchannel only
  passive     Enable LACP only if a LACP device is detected

ASW1(config-if-range)#channel-group 1 mode active
Creating a port-channel interface Port-channel 1
```

⚙️ EtherChannel Protocols

🔹 1. PAgP (Port Aggregation Protocol)

- Cisco proprietary 🟦
- Max: **8 links**

Modes:

Mode	Meaning
auto	passive
desirable	active

✓ Working:

- desirable + auto = ✅
- desirable + desirable = ✅
- auto + auto = ❌

🔹 2. LACP (Link Aggregation Control Protocol)

- Industry standard 🌍 (IEEE 802.3ad)
- Max: **16 links (8 active + 8 standby)**

Modes:

Mode	Meaning
active	active
passive	passive

✓ Working:

- active + passive = ✅
- active + active = ✅
- passive + passive = ❌

🔹 **3. Static EtherChannel**

- No protocol ❌
- Manual configuration

✔ Only works:

- on + on = ✅
- on + anything else = ❌

⚖️ Load Balancing in EtherChannel

🔹 **Based on "Flows"**

- A **flow = communication between two devices**

👉 Same flow → same cable
👉 Different flows → different cables

⚠️ **Why?**

To avoid **packet out-of-order problem**

🔧 **Load Balancing Methods**

- Source MAC
- Destination MAC
- Source + Destination MAC
- Source IP
- Destination IP
- Source + Destination IP

🔍 Check Load Balancing

show etherchannel load-balance

🔧 Change Method

conf t
port-channel load-balance src-dst-mac

🛠️ Configuration of EtherChannel

🔹 PAgP Configuration

interface range g0/0 - 3
channel-group 1 mode desirable

✔ Creates:

Port-channel 1

🔹 LACP Configuration

interface range g0/0 - 3
channel-group 1 mode active

🔹 Static Configuration

interface range g0/0 - 3
channel-group 1 mode on

⚠️ Important Rule

✔ Channel-group number:

- MUST match on same switch
- DOES NOT need to match on other switch

🔧 Manual Protocol Selection

channel-protocol lacp
channel-group 1 mode active

❌ If mismatch → ERROR

🔌 Configure Port-Channel Interface

interface port-channel 1
switchport trunk encapsulation dot1q
switchport mode trunk

📋 Running Config Example

- Physical interfaces auto-configured from port-channel

interface GigabitEthernet0/0
channel-group 1 mode active

⚠️ Important Conditions

All member interfaces MUST have:

✔ Same speed
✔ Same duplex
✔ Same VLAN
✔ Same trunk/access mode

❌ Otherwise → interface excluded

🔍 Verification Commands (VERY IMPORTANT)

✅ 1. Show Summary

show etherchannel summary

Output Meaning:

Symbol	Meaning
S	Layer 2
R	Layer 3
U	In use
P	Bundled
s	Suspended ❌
D	Down ❌

✔ Correct:

- L2 → SU
- L3 → RU

✅ 2. Port-Channel Info

show etherchannel port-channel

✅ 3. Trunk Check

show interfaces trunk

✅ 4. IP Check

show ip interface brief

✅ 5. Running Config

show running-config

✅ 6. Load Balance

show etherchannel load-balance

🌐 Layer 3 EtherChannel Configuration

interface range g0/0 - 3
no switchport
channel-group 1 mode active

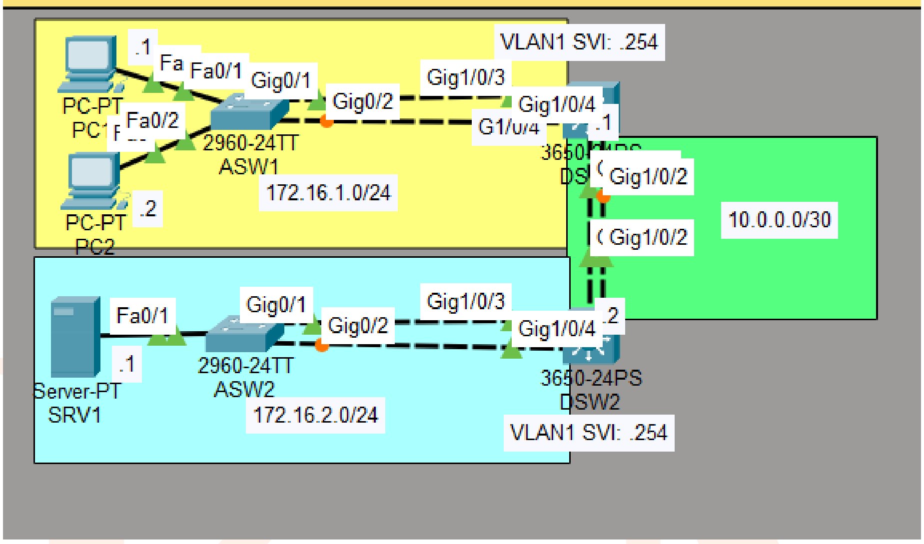
Then:

interface port-channel 1
ip address 10.0.0.1 255.255.255.252

✓ IP is assigned on **Port-channel**, not physical ports

🧪 EtherChannel LAB 🧠

🔹 Topology



- ASW1 ↔ DSW1 → LACP
- ASW2 ↔ DSW2 → PAgP
- DSW1 ↔ DSW2 → Layer 3 Static
- PCs → ASW1
- Server → ASW2

✅ 1. Configure Layer 2 EtherChannel between ASW1 and DSW1 using LACP.Configure it as a trunk.

🔹 On ASW1

```
conf t
interface range g0/1 - 2
switchport mode trunk
channel-group 1 mode active
exit
```

interface port-channel 1
switchport mode trunk
do wr

🔹 **On DSW1**

conf t
interface range g1/0/3 - 4
switchport mode trunk
channel-group 1 mode passive
exit

interface port-channel 1
switchport mode trunk
do wr

✔ **active + passive = LACP works**

✅ **2. Configure Layer 2 EtherChannel between ASW2 and DSW2 using PAgP. Configure it as a trunk.**

🔹 **On ASW2**

conf t
interface range g0/1 - 2
switchport mode trunk
channel-group 2 mode desirable
exit

interface port-channel 2
switchport mode trunk
do wr

◆ **On DSW2**

```
conf t
interface range g1/0/3 - 4
switchport mode trunk
channel-group 2 mode auto
exit

interface port-channel 2
switchport mode trunk
do wr
```

✔ **desirable + auto = PAgP works**

✅ **3. Configure Layer 3 EtherChannel between DSW1 and DSW2 using static EtherChannel.**

◆ **On DSW1**

```
conf t
interface range g1/0/1 - 2
no switchport
channel-group 3 mode on
exit

interface port-channel 3
no switchport
ip address 10.0.0.1 255.255.255.252
end
```

◆ **On DSW2**

```
conf t
interface range g1/0/1 - 2
no switchport
channel-group 3 mode on
exit
```

interface port-channel 3
no switchport
ip address 10.0.0.2 255.255.255.252
end

✔ Static EtherChannel → **on + on only works**

✅ 4. Configure routes to allow the PCs to reach SRV1.

- Server network behind DSW2
- PCs behind DSW1

🔹 **On DSW1**

ip route 172.168.1.0 255.255.255.0 10.0.0.2

🔹 **On DSW2**

ip route 172.16.2.0 255.255.255.0 10.0.0.1

✔ This allows both sides to communicate

✅ 5. What is the default EtherChannel load-balancing method used on each switch?

Check on any switch:

show etherchannel load-balance

✔ Default output usually:

src-dst-ip

👉 Meaning:
Source XOR Destination IP

✅ **6. Configure the switches to load-balance based on source and destination IP addresses.**

🔹 **On ALL switches**

```
conf t
port-channel load-balance src-dst-ip
end
```

Verify:

```
show etherchannel load-balance
```

🔍 Verification Commands (VERY IMPORTANT)

```
show etherchannel summary
show etherchannel port-channel
show interfaces trunk
show ip interface brief
show spanning-tree
```

Output flags:

- **SU** → Layer 2 + in use ✅
- **RU** → Layer 3 + in use ✅
- **P** → bundled port ✅
- **s** → suspended ❌

⚠️ Common Mistakes

❌ VLAN mismatch
❌ Trunk/Access mismatch
❌ Speed/Duplex mismatch
❌ Wrong mode combination
❌ Using "on" with LACP/PAgP

🧩 Quick Revision Table

Type	Protocol	Modes	Works
LACP	active/passive	active + passive	✅
PAgP	desirable/auto	desirable + auto	✅
Static	on	on + on	✅

🎯 Final Summary

✓ EtherChannel = Multiple links → One logical link
✓ Prevents STP blocking
✓ Improves speed + redundancy 🚀
✓ Load balancing using flows
✓ LACP (standard) > PAgP (Cisco only)

EtherChannel Summary 🚀

🧠 Core Concept

👉 EtherChannel = **Multiple physical links combined into ONE logical link**

✓ Multiple cables → ONE logical interface
✓ Multiple ports → ONE port
✓ STP treats it as **single link**

🎯 Why EtherChannel is Needed

❌ Without EtherChannel:

- STP blocks redundant links 😑
- Bandwidth waste
- Congestion (Oversubscription)

✅ With EtherChannel:

- All links ACTIVE 🔥
- Load balancing ⚖️
- No STP blocking
- High speed + redundancy 🚀

🧩 Types of EtherChannel

Type	Description
Layer 2	Works like switch port (VLAN, trunk)
Layer 3	Routed port with IP address

⚙️ Protocols

Protocol	Type	Key Point
LACP	Standard (IEEE 802.3ad)	Most used 🌍
PAgP	Cisco only	Old
Static	Manual	Risky

⚖️ Load Balancing

✓ Based on **flows (not packets)**
✓ Prevents packet disorder

Methods:

- MAC based
- IP based (most common)

⚠️ Important Conditions

All interfaces must have:
✓ Same speed
✓ Same duplex
✓ Same VLAN
✓ Same trunk/access mode

🔍 Verification Commands

show etherchannel summary
show interfaces trunk
show ip interface brief
show etherchannel load-balance

🎯 ✅ Conclusion 🎓

👉 EtherChannel is used to:

- Increase bandwidth 🚀
- Provide redundancy 🔁
- Avoid STP blocking ❌
- Improve performance ⚡

📌 **Best Practice:**
👉 Use **LACP (active/passive)** in real networks

👉 Final one-liner:
"EtherChannel combines multiple physical links into one logical link to increase bandwidth and provide redundancy."

🎤 Q & A

❓ Q1: What is EtherChannel?

✅ **Answer:**
EtherChannel is a technology that combines multiple physical interfaces into a single logical interface, allowing increased bandwidth and redundancy while preventing STP from blocking links.

❓ Q2: Why do we use EtherChannel?

✅ **Answer:**
We use EtherChannel to:

- Increase bandwidth 🚀
- Prevent STP blocking ❌
- Provide redundancy 🔁
- Improve network performance ⚡

❓ Q3: What is LAG?

✅ **Answer:**
LAG (Link Aggregation Group) is another name for EtherChannel.

❓ Q4: What problem does EtherChannel solve?

✅ **Answer:**
It solves:

- STP blocking issue
- Oversubscription problem
- Bandwidth limitation

⚙️ **PROTOCOL QUESTIONS**

❓ **Q5: Difference between LACP and PAgP?**

✅ **Answer:**

Feature	LACP	PAgP
Type	Standard	Cisco only
IEEE	802.3ad	No
Usage	Modern networks	Legacy

❓ **Q6: LACP modes?**

✅ **Answer:**

- active (initiates)
- passive (waits)

✓ active + passive = works
❌ passive + passive = fails

❓ **Q7: PAgP modes?**

✅ **Answer:**

- desirable (active)
- auto (passive)

✓ desirable + auto = works
❌ auto + auto = fails

❓ Q8: Static EtherChannel?

✅ **Answer:**

- Manual configuration
- No negotiation
  ✓ on + on works
  ❌ mismatch fails

❓ Q9: Configure LACP EtherChannel

✅ **Answer:**

interface range g0/0 - 3
channel-group 1 mode active

❓ Q10: Configure trunk on port-channel

interface port-channel 1
switchport mode trunk

❓ Q11: Layer 3 EtherChannel config

interface range g0/0 - 3
no switchport
channel-group 1 mode active

interface port-channel 1
ip address 10.0.0.1 255.255.255.252

❓ Q12: How to verify EtherChannel?

show etherchannel summary

✓ Look for:

- SU → Layer 2 working
- RU → Layer 3 working
- P → bundled

❓ Q13: How does load balancing work?

✅ **Answer:**

- Based on flows (not packets)
- Same flow → same link
- Different flows → different links

👉 Prevents packet reordering

❓ Q14: Why STP does not block EtherChannel?

✅ **Answer:**
Because STP sees EtherChannel as a single logical interface.

❓ Q15: Max links in LACP?

✅ **Answer:**

- 16 total
- 8 active + 8 standby

❓ **Q16: What happens if configuration mismatch?**

✅ **Answer:**

- Ports go to **suspended state**
- EtherChannel fails ❌

⚠️ **TROUBLESHOOTING QUESTIONS**

❓ **Q17: EtherChannel not forming – reasons?**

✅ **Answer:**
Common issues:

- VLAN mismatch ❌
- Trunk mismatch ❌
- Speed mismatch ❌
- Duplex mismatch ❌
- Wrong mode ❌

❓ **Q18: What is suspended port?**

✅ **Answer:**
Port is not added to EtherChannel due to mismatch configuration.

❓ **Q19: Command to check load balancing?**

show etherchannel load-balance

❓ **Q20: Change load balancing method**

port-channel load-balance src-dst-ip

🎯 SCENARIO-BASED QUESTIONS

❓ Q21: Why only one link used without EtherChannel?

✅ **Answer:**
Because STP blocks redundant links to prevent loops.

❓ Q22: What happens if one link fails in EtherChannel?

✅ **Answer:**

- Traffic shifts to remaining links
- No downtime 🔥

❓ Q23: Can we mix FastEthernet and Gigabit?

❌ **Answer:**
No, all interfaces must have same speed.

❓ Q24: Can we use different VLANs on member ports?

❌ No — must be identical

❓ Q25: Difference between Layer 2 and Layer 3 EtherChannel?

✅ **Answer:**

Feature	Layer 2	Layer 3
Works on	MAC	IP
VLAN support	Yes	No
IP address	No	Yes

❓ Q26: What is flow-based hashing?

✅ **Answer:**
Algorithm used to decide which link carries traffic.

❓ Q27: What is port-channel interface?

✅ **Answer:**
Logical interface representing all bundled physical ports.

❓ Q28: How to create EtherChannel between switches?

Steps:
1️⃣ Select interfaces
2️⃣ Configure mode
3️⃣ Create port-channel
4️⃣ Apply trunk

❓ Q29: How to verify trunk in EtherChannel?

show interfaces trunk

❓ Q30: What happens if one side is LACP and other is PAgP?

❌ EtherChannel will NOT form

🚀 One Liner

"Explain EtherChannel in simple terms"

💯 Answer:
👉 "EtherChannel combines multiple physical links into one logical link to increase bandwidth, provide redundancy, and prevent STP from blocking links."
